POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2019
Programme: BE | Full Marks: 100
Course: Computer Architecture | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the different evolutionary milestones in development of computer system. 7
   b) What are the different types of register used in computer system, explain with examples. 8
2. a) Write the algorithm and perform multiplication of 9*-3 using booth algorithm. 8
   b) Differentiate between micro programmed control and hardwired control unit? What are the techniques used to sequence microinstructions.
3. a) What is instruction cycle? Explain various sub cycle of instruction cycle with micro-instructions involved in each sub cycle.
   b) Define virtual memory. Explain address mapping using pages in virtual memory.
4. a) Explain how redundancy is obtained in RAID, with reference to any two RAID levels. What is the application of RAID?
   b) Explain with an example, how effective address is calculated in Direct, Register indirect, Relative and Index addressing modes. (Assume necessary data if you need)
5. a) What is mean by pipelining? Explain in brief about RISC pipelining?
   b) What is cache coherence problem? Explain methods of solving cache coherence problem.
6. a) What are the various interconnected structure used for multiprocessor system? Explain them.
   b) Briefly explain the different performance issues in Hardware.
7. Write short notes on: (Any two) 2
   a) Register Transfer language (HDL)
   b) Future trends in computer
   c) Instruction pipelining

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